

NeuroScan AI - Technology Stack & Tools Inventory

🖥️ Programming Languages

****Primary:****

- ****Python (v3.9+)****: The core backbone of the entire backend, AI processing, and server logic.
- ****JavaScript (ES6+)****: The engine for all frontend interactivity, 3D visualization, and client-side logic.
- ****HTML5****: The semantic structure of the application interface.
- ****CSS3****: The styling language used to create the advanced "Glassmorphism" and responsive design.

****Secondary / Scripting:****

- ****Batch Scripting**** (`.bat`): Used for Windows automation (starting servers, environment setup).
- ****SQL****: Used via SQLite for querying the local patient & report database.
- ****Markdown****: Used for all project documentation.

🛠️ Frameworks & Libraries (Big Tools)

Backend (AI & Server)

- ****FastAPI****: The high-performance web framework used to build the REST API endpoints.
- ****PyTorch****: The deep learning framework powering the ****EfficientNetV2**** feature extractor.
- ****Scikit-Learn****: The machine learning library used for the ****SVM (Support Vector Machine)**** classifier.
- ****TensorFlow / Keras****: Used for the secondary consensus models (Swin Transformer & CoAtNet).
- ****OpenCV (cv2)****: The computer vision library used for image preprocessing, thresholding, and advanced analytics (contours, sizing).
- ****NumPy****: The fundamental package for all numerical computation and matrix operations.
- ****Pillow (PIL)****: Python Imaging Library for basic image file handling and format conversion.

Frontend (UI & Viz)

- ****Three.js****: The powerful 3D library used to render the ****Holographic Brain**** and tumor visualizations in the browser.
- ****Chart.js****: Used for rendering the "Diagnosis Trends" and analytics graphs.

- ****html2pdf.js****: The tool that converts the HTML diagnostic report into a downloadable PDF file.

Small Utilities & Modules

- ****Uvicorn****: An ASGI web server implementation to run the FastAPI app.
- ****Joblib****: Used for efficient serializing/loading of the SVM model files.
- ****Timm (PyTorch Image Models)****: A library providing the pre-trained EfficientNet architecture.
- ****Pydantic****: Used for strict data validation and schema definition in the API.
- ****SQLAlchemy****: The ORM (Object Relational Mapper) used to interact with the database in Pythonic code.
- ****Bcrypt****: Used for secure password hashing and verification.
- ****Python-Multipart****: Required for handling file uploads (MRI images) in FastAPI.
- ****Google Fonts API****: Feeds the specific typography (Inter, Outfit, JetBrains Mono).
- ****UI Avatars API****: Generates dynamic user avatars based on names.

Databases & Storage

- ****SQLite****: A lightweight, serverless relational database engine used for storing persistent data (no external server required).
- ****Local File System****: Used for storing the raw MRI image uploads (`/assets/uploads`) and generated PDF reports (`/assets/reports`).

Development Environment

- ****VS Code****: The Integrated Development Environment (IDE) likely used for writing code.
- ****Virtualenv (venv)****: Used to create an isolated Python environment to manage dependencies.